

# Inequalities

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# 1 Inequalities

## I Triangle Inequality (三角不等式)

For any normed space  $V$  with norm  $\|\cdot\|$ :

$$\forall a, b \in V : \|a\| + \|b\| \geq \|a + b\| \geq |\|a\| - \|b\||.$$

## II Reverse Triangle Inequality (反三角不等式)

For any normed space  $V$  with norm  $\|\cdot\|$ :

$$\forall a, b \in V : \|a\| + \|b\| \geq \|a - b\| \geq |\|a\| - \|b\||.$$

## III Jensen's Inequality (詹森不等式)

Let  $(\Omega, \mu)$  be a probability space,  $g : \Omega \rightarrow \mathbb{R}$  be a real-valued  $\mu$ -integrable function, and  $\varphi : \mathbb{R} \rightarrow \mathbb{R}$  be a convex function. Then, Jensen's inequality states:

$$\int_{\Omega} \varphi \circ g(\omega) \, d\mu(\omega) \geq \varphi \left( \int_{\Omega} g(\omega) \, d\mu(\omega) \right)$$

*Proof.*

Since  $\varphi$  is convex, at each real number  $x$ , we have a non-empty set of subderivatives, which may be thought of as lines touching the graph of  $\varphi$  at  $x$ , but which are below the graph of  $\varphi$  at all points.

We define:

$$x_0 := \int_{\Omega} g \, d\mu.$$

Because of the existence of subderivatives for convex functions, we may choose  $a$  and  $b$  such that  $ax + b \leq \varphi(x)$  for all real  $x$  and  $ax_0 + b = \varphi(x_0)$ .

But then we have that for almost all  $\omega \in \Omega$ :

$$\varphi(g(\omega)) \geq ag(\omega) + b$$

Since we have a probability measure, the integral is monotone with  $\mu(\Omega) = 1$  so that

$$\begin{aligned} \int_{\Omega} \varphi \circ g(\omega) \, d\mu(\omega) &\geq \int_{\Omega} ag(\omega) + b \, d\mu(\omega) \\ &= a \int_{\Omega} g \, d\mu + b \int_{\Omega} d\mu \\ &= ax_0 + b = \varphi(x_0) \\ &= \varphi \left( \int_{\Omega} g \, d\mu \right). \end{aligned}$$

□

## IV AM-GM Inequality (算幾不等式)

$$\frac{\sum_{i=1}^n x_i}{n} \geq \left( \prod_{i=1}^n x_i \right)^{1/n}$$

*Proof.*

*Lemma:* Jensen's inequality.

Applying Jensen's inequality to the logarithm function, which is concave, and the arithmetic mean:

$$\log \left( \frac{\sum_{i=1}^n x_i}{n} \right) \geq \sum_{i=1}^n \frac{1}{n} \log(x_i) = \log \left( \left( \prod_{i=1}^n x_i \right)^{1/n} \right)$$

Exponentiating both sides gives the desired inequality:

$$\frac{\sum_{i=1}^n x_i}{n} \geq \left( \prod_{i=1}^n x_i \right)^{1/n}$$

□

## V Cauchy-Schwarz Inequality (柯西-施瓦茨不等式)

For any  $\mathbf{u}, \mathbf{v} \in \mathbb{C}^n$ ,

$$|\mathbf{u} \cdot \mathbf{v}| \leq |\mathbf{u}| |\mathbf{v}|$$

and

$$|\mathbf{u} \cdot \mathbf{v}| \leq |\mathbf{u}| |\mathbf{v}| \iff \mathbf{u} \parallel \mathbf{v},$$

where  $\mathbf{u} \cdot \mathbf{v}$  is the standard inner product defined as

$$\mathbf{u} \cdot \mathbf{v} = \sum_{i=1}^n u_i \bar{v}_i,$$

with  $\bar{v}_i$  denoting the complex conjugate of  $v_i$ .

*Proof.*

Consider the complex number  $z = \mathbf{u} \cdot \mathbf{v}$ .

Define the function

$$f(t) = |\mathbf{u} + t\mathbf{v}|^2$$

for some real number  $t$ . Then, we have

$$\begin{aligned} f(t) &= (\mathbf{u} + t\mathbf{v}) \cdot \overline{(\mathbf{u} + t\mathbf{v})} \\ &= (\mathbf{u} \cdot \bar{\mathbf{u}}) + t(\mathbf{u} \cdot \bar{\mathbf{v}}) + t(\mathbf{v} \cdot \bar{\mathbf{u}}) + t^2(\mathbf{v} \cdot \bar{\mathbf{v}}) \\ &= |\mathbf{u}|^2 + 2t\Re(\mathbf{u} \cdot \bar{\mathbf{v}}) + t^2|\mathbf{v}|^2. \end{aligned}$$

Since  $f(t) \geq 0$  for all  $t \in \mathbb{R}$ , the quadratic equation in  $t$  must have a non-positive discriminant. The discriminant of this quadratic is:

$$\begin{aligned} \Delta &= (2\Re(\mathbf{u} \cdot \bar{\mathbf{v}}))^2 - 4 \times |\mathbf{v}|^2 \times |\mathbf{u}|^2 \\ &= 4(\Re(\mathbf{u} \cdot \bar{\mathbf{v}}))^2 - 4|\mathbf{u}|^2|\mathbf{v}|^2. \end{aligned}$$

We require  $\Delta \leq 0$ . This implies:

$$(\Re(\mathbf{u} \cdot \bar{\mathbf{v}}))^2 \leq |\mathbf{u}|^2 |\mathbf{v}|^2.$$

Taking the square root of both sides and noting that  $|\mathbf{u} \cdot \mathbf{v}| \geq \Re(\mathbf{u} \cdot \bar{\mathbf{v}})$ :

$$|\mathbf{u} \cdot \mathbf{v}| \leq |\mathbf{u}| |\mathbf{v}|.$$

Equality  $|\mathbf{u} \cdot \mathbf{v}| = |\mathbf{u}| |\mathbf{v}|$  holds if and only if the discriminant  $\Delta = 0$ . This happens when the quadratic equation has a double root or equivalently, when  $\mathbf{u} + t\mathbf{v} = 0$  for some real  $t$ , implying  $\mathbf{u}$  is a scalar multiple of  $\mathbf{v}$ . Therefore,  $\mathbf{u}$  and  $\mathbf{v}$  are linearly dependent, meaning:

$$\mathbf{u} \parallel \mathbf{v}.$$

□

For any  $\mathbf{u}, \mathbf{v} \in \mathbb{R}^n$ ,

$$|\mathbf{u} \cdot \mathbf{v}| \leq |\mathbf{u}| |\mathbf{v}|$$

and

$$|\mathbf{u} \cdot \mathbf{v}| \leq |\mathbf{u}| |\mathbf{v}| \iff \mathbf{u} \parallel \mathbf{v},$$

where  $\mathbf{u} \cdot \mathbf{v}$  is the standard inner product defined as

$$\mathbf{u} \cdot \mathbf{v} = \sum_{i=1}^n u_i v_i.$$